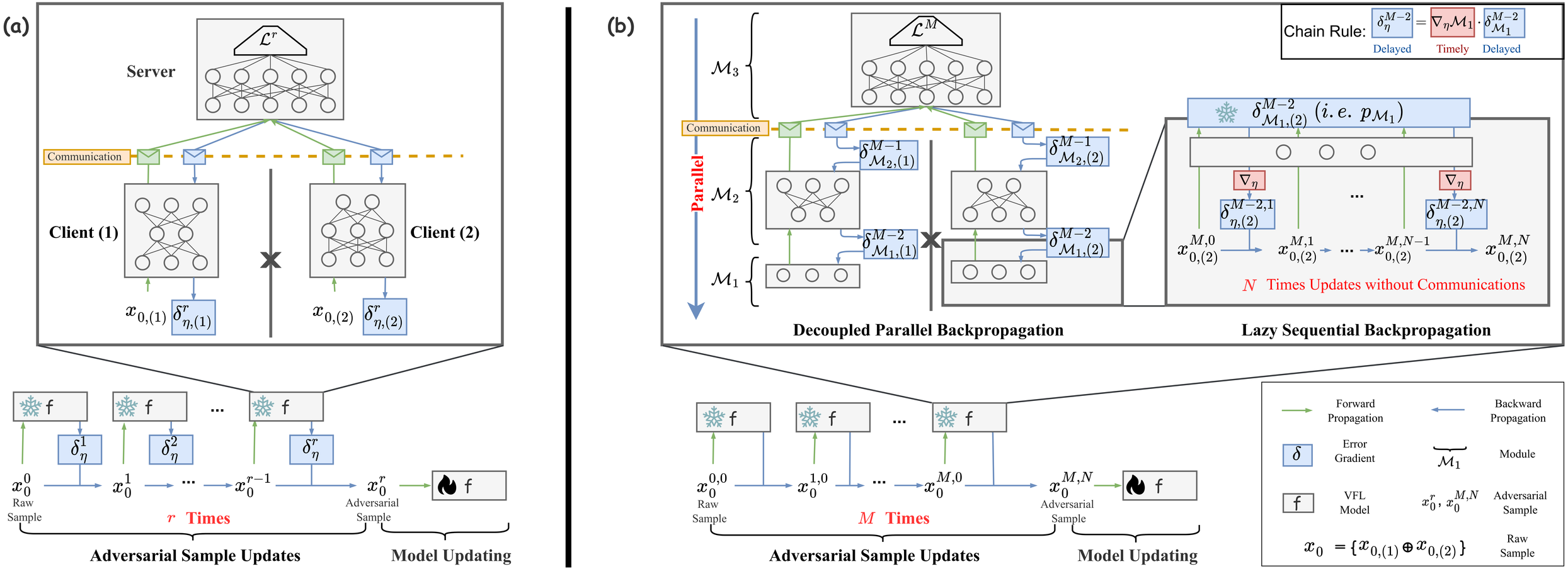
(a)
Server
$\mathcal{L}^r$
Communication
Client (1)
Client (2)
$x_{0,(1)}$ $\delta^r_{\eta,(1)}$
$x_{0,(2)}$ $\delta^r_{\eta,(2)}$
f
δ^1_η
δ^2_η
δ^r_η
x_0^0 Raw Sample
x_0^1
x_0^{r-1}
x_0^r Adversarial Sample
r Times
Adversarial Sample Updates
Model Updating
(b)
$\mathcal{L}^M$
$\mathcal{M}_3$
$\mathcal{M}_2$
$\mathcal{M}_1$
Parallel
Communication
$\delta^{M-1}_{\mathcal{M}_2,(1)}$
$\delta^{M-1}_{\mathcal{M}_2,(2)}$
$\delta^{M-2}_{\mathcal{M}_1,(1)}$
$\delta^{M-2}_{\mathcal{M}_1,(2)}$
Decoupled Parallel Backpropagation
Chain Rule: $\delta^{M-2}_\eta = \nabla_\eta \mathcal{M}_1 \cdot \delta^{M-2}_{\mathcal{M}_1}$
Delayed Timely Delayed
$\delta^{M-2}_{\mathcal{M}_1,(2)}$ (i.e. $p_{\mathcal{M}_1}$)
∇_η
$\delta^{M-2,1}_{\eta,(2)}$
$\delta^{M-2,N}_{\eta,(2)}$
$x^{M,0}_{0,(2)}$
$x^{M,1}_{0,(2)}$
$x^{M,N-1}_{0,(2)}$
$x^{M,N}_{0,(2)}$
N Times Updates without Communications
Lazy Sequential Backpropagation
$x_0^{0,0}$ Raw Sample
$x_0^{1,0}$
$x_0^{M,0}$
$x_0^{M,N}$ Adversarial Sample
M Times
Adversarial Sample Updates
Model Updating
Forward Propagation
Backward Propagation
Error Gradient
Module
VFL Model
x_0^r, $x_0^{M,N}$ Adversarial Sample
$x_0 = \{x_{0,(1)} \oplus x_{0,(2)}\}$ Raw Sample